Paper I – Basic Sciences in Urology (MCh Urology, Summer 2016)

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I-1. Discuss Iodinated contrast radiological agents and their complications.

Type: LAQ (25 marks)

Introduction

- Iodine (atomic number 53, K-edge 33 keV) absorbs X-rays strongly. All agents are built on a **tri-iodinated benzene ring**.
- They are filtered by the glomerulus, not secreted or reabsorbed, so they concentrate in urine. Half-life is about 2 hours with normal function; biliary excretion rises in renal failure.
- Uses: CT (computed tomography) urography, IVU (intravenous urography), pyelography, cystourethrography, angiography.

Classification

Class	Example	Osmolality (mOsm/kg)	Iodine : particles
High-osmolar ionic monomer	Diatrizoate	~1500–2000 (5–8 × plasma)	3 : 2
Low-osmolar non-ionic monomer	Iohexol, iopamidol	~600–850	3 : 1
Low-osmolar ionic dimer	Ioxaglate	~600	6 : 2
Iso-osmolar non-ionic dimer	Iodixanol	~290	6 : 1

- **Why it matters:** a higher ratio gives more opacity per particle. Fewer particles mean less fluid shift, pain and reactions. So non-ionic low- or iso-osmolar agents are standard intravascularly.

Complications

A. Hypersensitivity (allergic-like) reactions

- Mostly non-IgE (immunoglobulin E) direct mast-cell release: idiosyncratic, not dose-related. Severe reactions: 0.22% with ionic vs 0.04% with non-ionic agents (Katayama, 1990).

Type	Timing	Features
Immediate	Within about 1 hour (most within minutes)	Urticaria, angio-oedema, bronchospasm, hypotension; anaphylaxis if severe

Type	Timing	Features
Non-immediate	Hours to days	Maculopapular rash; T-cell mediated, milder

- Risk factors: previous reaction (strongest), asthma, atopy, severe cardiac disease, drug allergy, female sex. **Shellfish ("iodine") allergy is not a cross-reaction.**
- **Physiological reactions:** warmth, metallic taste, nausea, vasovagal hypotension with bradycardia, arrhythmia, seizures.

B. CA-AKI (contrast-associated acute kidney injury)

- **CA-AKI** = AKI within 48 hours of contrast, whatever the cause. **CI-AKI (contrast-induced AKI)** = the subset caused by contrast.
- KDIGO (Kidney Disease: Improving Global Outcomes) AKI: creatinine rise ≥0.3 mg/dL in 48 hours, ≥1.5 × baseline in 7 days, or urine <0.5 mL/kg/h for ≥6 hours.
- **Mechanism:** afferent vasoconstriction causes medullary hypoxia (ischaemic ATN (acute tubular necrosis)), plus direct tubular toxicity.
- **Risk factors:** CKD (chronic kidney disease), diabetes, age, anaemia, contrast volume, intra-arterial route.
- ACR (American College of Radiology)/NKF (National Kidney Foundation) 2020: risk was overstated; it is rare but real, mainly at eGFR (estimated glomerular filtration rate) <30 mL/min/1.73 m².

C. Others

- **Metformin:** lactic acidosis only if AKI develops. ACR: eGFR ≥30 and no AKI, continue. eGFR <30 or AKI, withhold 48 hours and restart when function is stable.
- **Thyroid:** iodine load can trigger hyperthyroidism (Jod-Basedow) in Graves' disease or nodular goitre. It blocks radio-iodine scans or therapy for several weeks.
- **Extravasation:** usually minor, but large power-injected volumes can rarely cause compartment syndrome or skin necrosis. Elevate the limb, apply compresses and observe. Get a surgical review if swelling increases, blisters form, or perfusion or sensation changes.
- **Also know:** iodide "mumps"; myasthenia flare; pregnancy (crosses placenta, not teratogenic); breastfeeding can continue; phaeochromocytoma safe with IV (intravenous) non-ionic agents; dialysis patients need no urgent dialysis.

Prevention and management

Premedication for high-risk patients (ACR 2023)

Regimen	Drugs
1	Prednisone 50 mg orally 13, 7 and 1 hour before + diphenhydramine 50 mg 1 hour before
2	Methylprednisolone 32 mg orally 12 and 2 hours before + diphenhydramine 50 mg 1 hour before

- First consider ultrasound, MRI (magnetic resonance imaging) or non-contrast CT. Breakthrough reactions still occur, so keep resuscitation drugs ready. Retrograde pyelography is safe even after anaphylaxis, because little contrast is absorbed.

Treating reactions

```
Stop injection → airway, oxygen, call for help
  Urticaria → observe ± H1-antihistamine (may progress)
  Bronchospasm → salbutamol 2 puffs (90 µg each), up to 3 times
  Anaphylaxis → intramuscular (IM) adrenaline 0.3–0.5 mg (1 mg/mL) every 5–15 min,
                max 1 mg; IV fluids
  Vasovagal → legs up, IV fluids, IV atropine
Observe ≥30 min; later allergy testing; record the reaction
```

Preventing CA-AKI

- Lowest dose of a low- or iso-osmolar agent; weight-based dosing; a low tube voltage (80 kVp, kilovolt peak) needs less contrast. Avoid repeat doses within 48 hours; stop NSAIDs (non-steroidal anti-inflammatory drugs).
- **IV isotonic saline** for eGFR <30 not on dialysis: the only proven measure.
- **PRESERVE (2018):** bicarbonate and N-acetylcysteine (NAC) gave no benefit. **AMACING (2017):** no prophylaxis was non-inferior to hydration at eGFR 30–59. Removing contrast by dialysis does not reduce risk.

Diagram to draw: tri-iodinated benzene ring, monomer versus dimer, on an osmolality scale.

Key points to remember:

- Ratio 1.5 (ionic), 3 (non-ionic monomer), 6 (iodixanol, 290 mOsm/kg).
- Anaphylaxis: IM adrenaline 0.3–0.5 mg, max 1 mg; shellfish allergy is not a risk.
- Premedication: prednisone 50 mg at 13-7-1 h or methylprednisolone 32 mg at 12-2 h, plus diphenhydramine.
- CA-AKI: saline if eGFR <30; bicarbonate and NAC useless; hold metformin only if eGFR <30 or AKI.

I-2. Discuss general principles of genitourinary infections, characteristics of common organisms and management of bacteremia and septic shock.

Type: LAQ (25 marks)

Definitions

- **UTI (urinary tract infection):** urothelial inflammation from bacterial invasion, usually with bacteriuria and pyuria. **Sterile pyuria** needs evaluation (tuberculosis, stone, tumour).
- **Uncomplicated:** healthy non-pregnant woman, normal tract. **Complicated:** men, pregnancy, obstruction, stones, catheter, neurogenic bladder, diabetes, immunosuppression.
- **Recurrent:** ≥2 in 6 months or ≥3 in 12 months. Unresolved = never cleared; reinfection = new organism; **bacterial persistence** = same organism from a focus (stone, prostate).
- **Prophylaxis** assumes a sterile tract; **suppression** controls a focus that cannot be eradicated.

Pathogenesis

- **Routes:** ascending from bowel flora (commonest); haematogenous (*S. aureus*, *Candida*); lymphatic (rare).
- **Virulence (uropathogenic *E. coli*): type 1 pili** (mannose-sensitive) bind bladder uroplakins and cause cystitis. **P pili** (mannose-resistant) bind renal glycolipids and are found in about 90% of pyelonephritis strains. Haemolysin damages tissue. Bacteria invade umbrella cells and form intracellular communities; quiescent reservoirs cause recurrence.
- **Principle:** a strong host is infected only by virulent strains; a compromised host (obstruction, catheter) by almost any organism.
- **Host defences:** voiding washout, lactobacilli, urine osmolality and pH, uromodulin (Tamm–Horsfall protein), urothelial exfoliation.
- **Risk factors: obstruction** (stasis impairs defences and turns UTI life-threatening), VUR (vesicoureteral reflux; with infection it scars the growing kidney), stones, catheters, diabetes, pregnancy, spermicides.
- **Diagnosis:** dipstick (nitrite, leucocyte esterase) rules out better than it rules in. Culture is the gold standard: $\geq 10^5$ CFU (colony-forming units)/mL (Kass); $\geq 10^2$ in a symptomatic woman or a catheter specimen. Image (ultrasound, CT (computed tomography)) in complicated infection or fever beyond 72 hours.

Characteristics of common organisms

Organism	Key features
E. coli	~85% of community and ~50% of hospital UTI; ST131 clone is ESBL (extended-spectrum β-lactamase) and quinolone resistant
Proteus	Urease splits urea into ammonia: alkaline urine (pH >7), **struvite stones**, persistence in stones
Klebsiella	Capsule; urease; ESBL and KPC (*K. pneumoniae* carbapenemase) producers
Pseudomonas	Nosocomial, catheter biofilm; needs ceftazidime, piperacillin, aminoglycoside, fluoroquinolone or meropenem
Enterococcus	**Always resistant to cephalosporins**; amoxicillin, nitrofurantoin; VRE (vancomycin-resistant)
S. saprophyticus	~10% of cystitis in young sexually active women

Organism	Key features
S. aureus	Haematogenous renal abscess; in urine, look for bacteraemia
Anaerobes, *Candida*	Abscesses, fistulae; *Candida* with catheters and diabetes

- **Also know:** emerging *Aerococcus urinae*; *Ureaplasma* and *Mycoplasma* with negative routine cultures; *Chlamydia* and tuberculosis do not grow on routine culture (sterile pyuria).
- **Antibiotic principles (see 2015 I-3f):** urine levels matter in cystitis, tissue levels in pyelonephritis and bacteraemia. Use a narrow spectrum and short course; adjust for renal function; **drain obstruction**.

Bacteraemia, sepsis and septic shock

Sepsis-3 (2016)

Term	Definition
Sepsis	Life-threatening organ dysfunction from a dysregulated host response to infection: **SOFA (Sequential Organ Failure Assessment) rise ≥2**
qSOFA (quick SOFA)	Respiratory rate ≥22, altered mentation, systolic blood pressure ≤100 mmHg; ≥2 = high risk (prompt only)
Septic shock	Vasopressor needed for MAP (mean arterial pressure) ≥65 mmHg **and** lactate >2 mmol/L despite fluids; mortality >40%

- SIRS (systemic inflammatory response syndrome) criteria were dropped as non-specific.
- **Pathophysiology:** endotoxin binds TLR4 (Toll-like receptor 4). Cytokines (TNF-α (tumour necrosis factor-α), interleukins IL-1 and IL-6) and nitric oxide cause vasodilatation, capillary leak, myocardial depression and DIC (disseminated intravascular coagulation), then organ failure.
- **Urological triggers:** infected obstructed kidney, PCNL (percutaneous nephrolithotomy), high-pressure ureteroscopy, prostate biopsy, TURP (transurethral resection of the prostate), infected stones, emphysematous pyelonephritis, abscess.

Management (Surviving Sepsis Campaign: hour-1 bundle 2018, guidelines 2021)

```
Lactate → blood cultures ×2 + urine culture
→ IV (intravenous) broad-spectrum antibiotics within 1 hour
  (shock or probable sepsis; within 3 h if sepsis only possible)
→ 30 mL/kg crystalloid if hypotensive or lactate ≥4 mmol/L
→ noradrenaline first line for MAP ≥65; add vasopressin
→ SOURCE CONTROL: stent or nephrostomy (equally effective),
  drain abscess, change catheter; stone surgery later
→ refractory shock: hydrocortisone 200 mg/day; ICU (intensive care unit)
→ de-escalate by culture at 48–72 h
```

- **Empirical antibiotics:** by local resistance: third-generation cephalosporin or piperacillin–tazobactam ± aminoglycoside; carbapenem if ESBL risk.
- **Duration:** usually 7–10 days; longer with an undrained focus or *S. aureus* bacteraemia. Step down to oral therapy once afebrile and stable.
- **Support:** oxygen, ventilation, glucose control, transfusion below haemoglobin 7 g/dL, thromboprophylaxis. Watch for AKI (acute kidney injury), ARDS (acute respiratory distress syndrome) and DIC.

Diagram to draw: pathway from *E. coli* adherence to bacteraemia and cytokine-driven shock.

Key points to remember:

- *E. coli* ~85% community UTI; P pili cause pyelonephritis; *Proteus* causes struvite; *Enterococcus* resists cephalosporins.
- Sepsis = SOFA ≥2; shock = vasopressor for MAP ≥65 plus lactate >2.
- Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline.
- Infected obstructed kidney: drain urgently; no definitive stone surgery in sepsis.

I-3a. External Sphincter Dyssynergia

Type: SAQ (10 marks)

Definition

- **DESD (detrusor–external sphincter dyssynergia):** the striated sphincter contracts involuntarily during detrusor contraction, causing functional obstruction, high pressure and poor emptying.
- **It requires a lesion between the pons and the sacral cord:** suprasacral spinal cord injury, MS (multiple sclerosis), transverse myelitis, myelomeningocele. Brain lesions (stroke, Parkinson's) do not cause it.
- Lesions above T6 may add bladder-neck (smooth sphincter) dyssynergia and autonomic dysreflexia.

Blaivas classification

Type	EMG (electromyography) pattern	Lesion
1	Activity rises with detrusor pressure, then relaxes suddenly at peak	Incomplete
2	Sporadic contractions throughout voiding	Complete
3	Crescendo–decrescendo contraction; obstruction throughout	Complete

- Alternative: intermittent versus continuous (continuous means a complete lesion). Type has no relation to lesion level.

Diagnosis

- **Pressure–flow study with EMG:** EMG rises during the detrusor contraction, with high pressure and interrupted flow. Needle EMG is more reliable than surface.
- **VUDS (videourodynamics):** narrowed membranous urethra with dilated posterior urethra; shows VUR (vesicoureteral reflux).
- **Exclude:** guarding reflex (normal); **pseudodyssynergia** (straining or suppressing a contraction); dysfunctional voiding (learned, no neurological lesion; Hinman syndrome in children).

Significance

- Worse when complete, continuous and **in men**: about half of untreated men get VUR, hydronephrosis, stones or urosepsis.
- DLPP (detrusor leak point pressure) ≥40 cmH_2O endangers the kidneys.

Management

- **First line:** CIC (clean intermittent catheterisation) plus antimuscarinic or β3-agonist; then intradetrusor botulinum toxin.
- **CIC impossible (men):** sphincterotomy or sphincteric botulinum toxin, with condom drainage. The goal is DLPP <40; repeat procedures are common.
- Others: suprapubic catheter, Brindley stimulator with deafferentation, diversion. Urethral stents are abandoned; oral drugs disappoint.
- **Also know:** α-blockers for bladder-neck dyssynergia; lifelong upper-tract surveillance.

Diagram to draw: urodynamic trace with rising detrusor pressure (Pdet), EMG bursts and interrupted flow.

Key points to remember:

- Lesion between pons and sacral cord; never brain lesions.
- Blaivas 1 incomplete; 2 and 3 complete.
- Worst in men; DLPP ≥40 cmH_2O hostile.
- CIC plus detrusor relaxation first; sphincterotomy if CIC impossible.

I-3b. Embryology of Posterior Urethral Valves.

Type: SAQ (10 marks)

Introduction

- PUV (posterior urethral valves): commonest cause of bladder outlet obstruction in boys (1.4–2.2 per 10,000 births). Morgagni described them (1769); Hugh Hampton Young named and classified them (1919).

Normal development

- Weeks 5–6: the **urorectal septum** divides the cloaca into the anorectal canal and the **urogenital sinus**.
- The **Wolffian (mesonephric) ducts** join the cloaca just before division. Their orifices normally migrate to the **verumontanum**, leaving small folds (plicae colliculi).
- The **urogenital membrane** normally disappears completely.
- Fetal urine starts at 10–14 weeks; after 16 weeks amniotic fluid is mainly fetal urine. So obstruction in this period damages bladder, kidneys and lungs.

Young types and their origin

Type	Anatomy	Embryology
I (~95%)	Leaflets from the verumontanum fuse anteriorly just proximal to the external sphincter	Abnormal insertion of the Wolffian duct ends into the anterolateral cloacal wall: hypertrophied inferior urethral crest
II	Folds from verumontanum to bladder neck	Not real (Stephens 1955): secondary bladder-neck hypertrophy
III (5–10%)	Annular diaphragm with a central hole	**Persistent urogenital membrane**

Other theories

- Hypertrophied mucosal folds (oldest); cloacal remnants; Wolffian–Müllerian maldevelopment.
- **COPUM (congenital obstructing posterior urethral membrane, Dewan):** one oblique membrane, torn by the first catheter into apparent type I leaflets. Plausible but unproven.
- **Cobb's collar (Moormann's ring):** a congenital ring-like narrowing of the proximal bulbar urethra, just distal to the external sphincter; some authors group it with type III valves.
- **Genetics:** mostly sporadic; chromosome 11 duplications; copy number variants (worse renal outcome); TBX5, PTK7.

Consequences

- Thick bladder, dilated posterior urethra (**keyhole sign**), hydroureteronephrosis, **renal dysplasia**, oligohydramnios, pulmonary hypoplasia. Pop-offs (VUR (vesicoureteral reflux), urinoma, ascites) relieve pressure.

Diagram to draw: posterior urethra with type I leaflets versus a type III membrane.

Key points to remember:

- Type I (95%): abnormal Wolffian duct insertion; type III: persistent urogenital membrane; type II not real.
- Urorectal septum divides the cloaca at weeks 5–6.
- COPUM: one membrane torn into leaflets by catheterisation.
- Dysplasia is fixed in utero; ablation cannot reverse it.

I-3c. Drugs associated with Acute Renal Failure.

Type: SAQ (10 marks)

Introduction

- AKI (acute kidney injury), KDIGO (Kidney Disease: Improving Global Outcomes): creatinine rise ≥0.3 mg/dL in 48 hours or ≥1.5 × baseline in 7 days, or urine <0.5 mL/kg/h for 6 hours.

Classification by mechanism

Mechanism	Drugs	How
Haemodynamic (prerenal)	**NSAIDs** (non-steroidal anti-inflammatory drugs)	Block prostaglandin afferent dilatation
	ACE (angiotensin-converting enzyme) inhibitors, ARBs (angiotensin receptor blockers)	Remove efferent constriction that holds GFR (glomerular filtration rate)
	Ciclosporin, tacrolimus, contrast; diuretics	Afferent vasoconstriction; volume loss
ATN (acute tubular necrosis)	**Aminoglycosides**, amphotericin B, vancomycin, cisplatin, ifosfamide, contrast	Direct tubular toxicity
AIN (acute interstitial nephritis)	**β-lactams**, sulfonamides, quinolones, rifampicin, NSAIDs, **PPIs (proton pump inhibitors)**, allopurinol, **checkpoint inhibitors**	Cell-mediated hypersensitivity
Crystals	Aciclovir, methotrexate, indinavir, sulfadiazine; tumour lysis	Intratubular obstruction
Microangiopathy	Gemcitabine, VEGF (vascular endothelial growth factor) inhibitors	Endothelial injury
Rhabdomyolysis	Statins (with fibrates)	Myoglobin toxicity, precipitation

- **"Triple whammy":** NSAID + ACE inhibitor/ARB + diuretic in a dehydrated patient.
- **False creatinine rise:** trimethoprim and cimetidine block creatinine secretion.

Recognition

Type	Urine
Prerenal	FeNa (fractional excretion of sodium) <1%, bland
ATN	FeNa >1%, muddy-brown casts
AIN	Sterile pyuria, white cell casts, ± rash, fever, eosinophilia

- AIN appears days to weeks after the drug starts; it is the commonest biopsy lesion with checkpoint inhibitors. Urine eosinophils are unreliable.

Prevention and management

- Identify risk (CKD (chronic kidney disease), age, diabetes). Dose by eGFR (estimated GFR); monitor drug levels; once-daily aminoglycoside.
- Hydrate before contrast or cisplatin; alkalinise for methotrexate.
- **Stop the culprit**, withhold NSAIDs and ACE inhibitors, restore volume. **Early steroids for AIN** (only about half of antibiotic AIN recovers fully).
- Dialysis for refractory AEIOU: acidosis, electrolytes, intoxication, overload, uraemia.

Diagram to draw: glomerulus with afferent (prostaglandin) and efferent (angiotensin II) control.

Key points to remember:

- NSAIDs block afferent dilatation; ACE inhibitors/ARBs block efferent constriction.
- ATN: aminoglycosides, amphotericin, cisplatin, contrast.
- AIN: β-lactams, PPIs, NSAIDs, checkpoint inhibitors; stop and give steroids.
- Crystals: aciclovir, methotrexate, indinavir.

I-3d. Anomalies detected on Prenatal scan between 16th and 20th weeks of gestation

Type: SAQ (10 marks)

Introduction

- The **anomaly scan is done at 18–20 weeks** (ACOG (American College of Obstetricians and Gynecologists)); by 20 weeks most urological anomalies are visible.
- **Why this window:** after about 16 weeks amniotic fluid is mainly fetal urine, so **oligohydramnios reflects renal failure or outlet obstruction**. The bladder should be seen filling and emptying.
- Survey kidneys, collecting system, ureters, bladder, genitalia, amniotic fluid, abdominal wall and spine. AFI (amniotic fluid index): normal 8–18 cm; oligohydramnios <5–6 cm.

UTD (urinary tract dilation) risk, 16–27 weeks

Parameter	A1 (low)	A2–3 (any one)
APRPD (anteroposterior renal pelvic diameter)	4 to <7 mm	≥7 mm
Calyces	Central or none	Peripheral
Parenchyma, ureter, bladder	Normal	Abnormal
Oligohydramnios	No	Unexplained

- A1: rescan after 32 weeks. A2–3: scan every 4 weeks. Both: postnatal scan after 48 hours.
- Causes: transient (50–70%), PUJ (pelviureteric junction) obstruction, VUR (vesicoureteral reflux), megaureter, duplex, LUTO (lower urinary tract obstruction).

Anomalies and signs

Anomaly	Signs
PUV (posterior urethral valves)	Male; thick bladder; **keyhole sign**; bilateral hydroureteronephrosis; oligohydramnios
Bilateral renal agenesis	**Anhydramnios after 16 weeks**, no bladder, no renal arteries: lethal Potter sequence
MCDK (multicystic dysplastic kidney)	Non-communicating cysts, no central pelvis
ARPKD (autosomal recessive polycystic kidney disease)	Large, homogeneously **bright** kidneys; oligohydramnios
Bladder exstrophy	**Bladder never seen**, low cord insertion, lower abdominal mass, wide pubis
Prune belly	Thin-walled large bladder, empty scrotum
Myelomeningocele	Spinal defect, lemon and banana signs

- **Also know:** duplex kidney with ureterocele (intravesical cyst); cloacal exstrophy (adds omphalocele, spinal defects); hydrocolpos; ambiguous genitalia; renal ectopia.

Implications

- Severe LUTO: fetal MRI (magnetic resonance imaging), urine biochemistry, vesicoamniotic shunt or fetal cystoscopy at 18–24 weeks. **PLUTO** (shunt trial): better survival, not renal function. Plan delivery with a multidisciplinary team.

Diagram to draw: fetal keyhole sign.

Key points to remember:

- Anomaly scan 18–20 weeks; after 16 weeks amniotic fluid = fetal urine.
- UTD A2–3 = APRPD ≥7 mm or any abnormal feature.
- Keyhole + oligohydramnios = PUV; absent bladder = exstrophy or agenesis.
- ARPKD large bright kidneys; MCDK non-communicating cysts.

I-3e. Foley's catheter.

Type: SAQ (10 marks)

History

- **Frederic Foley**, a Boston urologist, developed the modern balloon catheter (Campbell: 1929; published in the 1930s) for **haemostasis after prostatectomy**. Reybard (1855) had made the first self-retaining one.

Design

- **Fr (French) = 3 × diameter in mm.** Adults usually 14–16 Fr; clot retention 20–24 Fr three-way. Newborn 6 Fr.
- Normal calibre: meatus 24 Fr (narrowest), bladder neck 28 Fr, prostatic urethra 32 Fr, female urethra 22 Fr.
- **Balloon:** usually 10 mL; larger (30 mL) for traction. Use **sterile water**; volume is printed on the port.
- **Lumens:** two-way (drainage + balloon) is routine; three-way adds irrigation for haematuria. Each lumen reduces drainage size.
- **Tips:** straight; **coudé** (curved, for prostatic enlargement); **Councill** (open, over a wire).
- **Materials:** latex (cheap, allergy); silicone (long term, less urethritis and encrustation); coatings (silver, antibiotic) reduce bacteriuria but not clinical infection.

Indications and contraindications

- **Indications:** retention, clot retention, output monitoring, long surgery, immobility, after urological surgery.
- **Absolute contraindication:** suspected urethral injury (blood at meatus, pelvic fracture). Do an RUG (retrograde urethrogram) first.
- **Relative:** AUS (artificial urinary sphincter): deactivate, then 12–14 Fr; stricture.

Technique

- Aseptic; 10–15 mL lignocaine gel. In men, advance **to the hub** and see urine before inflating. Replace the foreskin.
- **Difficult:** 18 Fr coudé → guidewire → cystoscopy, Councill over wire → suprapubic catheter.

Complications

- **CAUTI (catheter-associated urinary tract infection):** 75% of hospital UTIs; bacteriuria 5–10% per catheter-day; **biofilm** forms within minutes.
- Trauma (about 0.3% of inpatients): false passage, balloon inflated in the urethra. Also stricture, traumatic hypospadias, encrustation, paraphimosis, non-deflating balloon (puncture with a wire).

CAUTI prevention

- Review daily and remove early; closed dependent system; clean meatus with water; culture only if symptomatic. Treat only symptomatic infection and **change the catheter**.

Diagram to draw: cross-section of a three-way Foley.

Key points to remember:

- Foley balloon catheter: designed for post-prostatectomy haemostasis.
- Fr = 3 × diameter (mm); inflate with sterile water.
- Suspected urethral injury → RUG first; AUS → deactivate, 12–14 Fr.
- CAUTI 5–10% per day; remove early; change the catheter when treating.

I-3f. Andrew Novick.

Type: SAQ (10 marks)

Assumed meaning: Andrew C. Novick and the urological work linked to his name, mainly nephron-sparing surgery.

Who he was

- **Andrew C. Novick** was a **Cleveland Clinic** urologist and long-time chairman of its urology department. He was a world authority on renal surgery, an editor of *Campbell-Walsh Urology* (9th edition), and died in 2008.

Contributions

1. **Nephron-sparing surgery (NSS) / partial nephrectomy (PN):** he established its technique, indications and long-term cancer control. His group showed tumours ≤4 cm did as well with NSS as with radical nephrectomy, supporting the T1a/T1b split.
2. **Renovascular surgery:** aortorenal, splenorenal and hepatorenal bypass; **bench (ex vivo) surgery with autotransplantation** for complex lesions.
3. **Renal transplantation** surgery.

Indications for PN

Category	Situations
Imperative	Bilateral RCC (renal cell carcinoma); solitary functioning kidney
Relative	Contralateral kidney threatened (stones, stenosis, diabetes, hypertension); hereditary RCC
Elective	cT1a (≤4 cm) with a normal other kidney; now larger selected tumours

- **Relative contraindications:** <20% of nephron mass left; pedicle encasement; venous thrombus.

Principles of NSS

1. Three-dimensional CT (computed tomography) maps tumour, vessels and collecting system.
2. **Early hilar control.** Keep warm ischaemia under about 25–30 minutes; if longer, use **ice-slush hypothermia** and mannitol.
3. Excise with a **negative margin** (width immaterial); intraoperative ultrasound for endophytic tumours.
4. **Watertight collecting-system closure**, vessel ligation, capsular renorrhaphy.
5. Drain for urine leak.

- Function afterwards depends on parenchymal **quality and quantity preserved**; ischaemia matters less. At least 20–30% of one kidney is needed to avoid dialysis.

Legacy

- PN is now the AUA (American Urological Association) and EAU (European Association of Urology) standard for cT1a tumours.

Diagram to draw: PN steps: clamp, ice slush, excision, closure, renorrhaphy.

Key points to remember:

- Novick (Cleveland Clinic): nephron-sparing, renovascular and bench surgery.
- PN indications: imperative, relative, elective (cT1a ≤4 cm).
- Warm ischaemia under about 25–30 minutes; ice slush if longer.
- Negative margin is enough; keep ≥20–30% of one kidney.

Rapid Revision Sheet – Paper I

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

I-1. Discuss Iodinated contrast radiological agents and their complications. (LAQ)

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- Anaphylaxis: IM adrenaline 0.3–0.5 mg, max 1 mg; shellfish allergy is not a risk.
- Premedication: prednisone 50 mg at 13-7-1 h or methylprednisolone 32 mg at 12-2 h, plus diphenhydramine.
- CA-AKI: saline if eGFR <30; bicarbonate and NAC useless; hold metformin only if eGFR <30 or AKI.

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- *E. coli* ~85% community UTI; P pili cause pyelonephritis; *Proteus* causes struvite; *Enterococcus* resists cephalosporins.
- Sepsis = SOFA ≥2; shock = vasopressor for MAP ≥65 plus lactate >2.
- Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline.
- Infected obstructed kidney: drain urgently; no definitive stone surgery in sepsis.

I-3a. External Sphincter Dyssynergia (SAQ)

- Lesion between pons and sacral cord; never brain lesions.
- Blaivas 1 incomplete; 2 and 3 complete.
- Worst in men; DLPP ≥40 cmH_2O hostile.
- CIC plus detrusor relaxation first; sphincterotomy if CIC impossible.

I-3b. Embryology of Posterior Urethral Valves. (SAQ)

- Type I (95%): abnormal Wolffian duct insertion; type III: persistent urogenital membrane; type II not real.
- Urorectal septum divides the cloaca at weeks 5–6.
- COPUM: one membrane torn into leaflets by catheterisation.
- Dysplasia is fixed in utero; ablation cannot reverse it.

I-3c. Drugs associated with Acute Renal Failure. (SAQ)

- NSAIDs block afferent dilatation; ACE inhibitors/ARBs block efferent constriction.
- ATN: aminoglycosides, amphotericin, cisplatin, contrast.
- AIN: β-lactams, PPIs, NSAIDs, checkpoint inhibitors; stop and give steroids.
- Crystals: aciclovir, methotrexate, indinavir.

I-3d. Anomalies detected on Prenatal scan between 16th and 20th weeks of gestation (SAQ)

- Anomaly scan 18–20 weeks; after 16 weeks amniotic fluid = fetal urine.
- UTD A2–3 = APRPD ≥7 mm or any abnormal feature.
- Keyhole + oligohydramnios = PUV; absent bladder = exstrophy or agenesis.
- ARPKD large bright kidneys; MCDK non-communicating cysts.

I-3e. Foley's catheter. (SAQ)

- Foley balloon catheter: designed for post-prostatectomy haemostasis.
- Fr = 3 × diameter (mm); inflate with sterile water.
- Suspected urethral injury → RUG first; AUS → deactivate, 12–14 Fr.
- CAUTI 5–10% per day; remove early; change the catheter when treating.

I-3f. Andrew Novick. (SAQ)

- Novick (Cleveland Clinic): nephron-sparing, renovascular and bench surgery.
- PN indications: imperative, relative, elective (cT1a ≤4 cm).
- Warm ischaemia under about 25–30 minutes; ice slush if longer.
- Negative margin is enough; keep ≥20–30% of one kidney.